



Semantics and Information Extraction

2019–20

Ion Androutsopoulos

<http://www.aueb.gr/users/ion/>

These slides are partly based on material from the books:

- “Speech and Language Processing” by D. Jurafsky and J.H. Martin, 2nd edition, Prentice Hall, 2009 and 3rd edition (in preparation, <https://web.stanford.edu/~jurafsky/slp3/>).
- “Artificial Intelligence – A Modern Approach” by S. Russel and P. Norvig, 2nd edition, Prentice Hall, 2003.

Contents

- **Semantic parsing:**
 - Representing the semantics of sentences in First-Order Predicate Logic (FOPL).
 - Mapping sentences to FOPL using λ -calculus.
- **Lexical semantics:**
 - Lexical semantic relations, WordNet, FrameNet, thematic roles, selectional restrictions.
- **Information extraction:**
 - Preprocessing and pattern-matching approaches.
 - Supervised relation extraction.
 - Unsupervised relation extraction.

DCG for simple arithmetic language

digit --> [zero].

digit --> [one].

...

digit --> [nine].

expression --> digit.

expression --> [open], expression, [plus], expression, [close].

expression --> [open], expression, [minus], expression, [close].

expression --> [open], expression, [star], expression, [close].

expression --> [open], expression, [slash], expression, [close].

open open two plus four close slash
open four minus one close close

$((2 + 4) / (4 - 1))$

➤ phrase(expression, [open, open, two, plus, four, close, slash,
open, four, minus, one, close, close]).

Yes.

Semantic parsing for the arithmetic language

digit(0) --> [zero].

digit(1) --> [one].

...

digit(9) --> [nine].

Inside braces we write additional constraints that need to be satisfied for the rule to be used. Here 'is' assigns the result of $X1 + X2$ to X . ('=' denotes unification in Prolog.)

expression(X) --> digit(X).

expression(X) --> [open], expression($X1$), [plus], expression($X2$),
[close], { X is $X1 + X2$ }.

expression(X) --> [open], expression($X1$), [minus],
expression($X2$), [close], { X is $X1 - X2$ }.

...

➤ phrase(expression(X), [open, open, two, plus, four, close, slash,
open, four, minus, one, close, close]).

$X = 2$.

Examples of formulae in First-Order Predicate Logic

- *All cats like milk.*

$$\forall x (\text{IsCat}(x) \Rightarrow \text{Likes}(x, \text{Milk}))$$

Optional for
the TA
course

- *There is a cat that likes milk.*

$$\exists x (\text{IsCat}(x) \wedge \text{Likes}(x, \text{Milk}))$$

- Attention: $\exists x (\text{IsCat}(x) \Rightarrow \text{Likes}(x, \text{Milk}))$ says “There is an x that: (i) is not a cat; or (ii) if it is a cat, it likes milk”.

- *Psita likes all dogs.*

$$\forall x (\text{IsDog}(x) \Rightarrow \text{Likes}(\text{Psita}, x))$$

- Attention: $\forall x (\text{IsDog}(x) \wedge \text{Likes}(\text{Psita}, x))$ says “Everything is a dog and Psita likes it”.

- *There is a cat that likes all dogs.*

$$\exists x (\text{IsCat}(x) \wedge \forall y (\text{IsDog}(y) \Rightarrow \text{Likes}(x, y)))$$

Examples of formulae in First-Order Predicate Logic

Optional for
the TA
course

- *Milos dislikes all cats.*

$$\forall x (\text{IsCat}(x) \Rightarrow \neg \text{Likes}(\text{Milos}, x))$$

- *All dogs dislike all cats.*

$$\forall x (\text{IsDog}(x) \Rightarrow \forall y (\text{IsCat}(y) \Rightarrow \neg \text{Likes}(x, y)))$$

or:

$$\forall x \forall y ((\text{IsDog}(x) \wedge \text{IsCat}(y)) \Rightarrow \neg \text{Likes}(x, y))$$

- *Every person likes his/her father.*

$$\forall x \forall y ((\text{IsPerson}(x) \wedge \text{IsFatherOf}(y, x)) \Rightarrow \text{Likes}(x, y))$$

$$\eta: \forall x (\text{IsPerson}(x) \Rightarrow \text{Likes}(x, \mathbf{FatherOf}(x)))$$

Examples of formulae in First-Order Predicate Logic

Optional for the
TA course

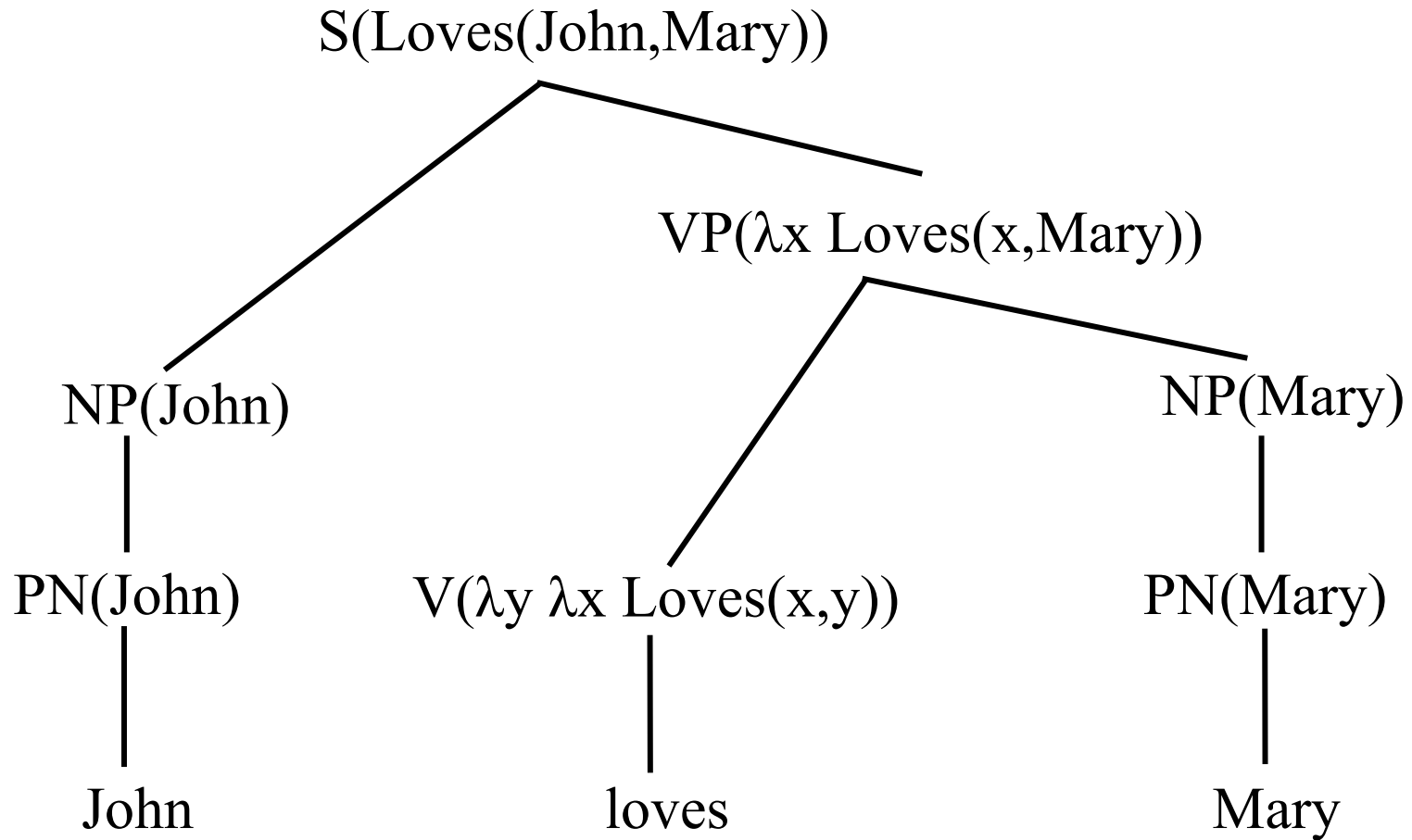
- *Every dog that barks is afraid of a (possibly different) cat.*

$$\forall x ((\text{IsDog}(x) \wedge \text{Barks}(x)) \Rightarrow \\ \exists y (\text{IsCat}(y) \wedge \text{IsAfraidOf}(x, y)))$$

- *Every cat likes exactly one (possibly different) dog.*

$$\forall y (\text{IsCat}(y) \Rightarrow \\ \exists x (\text{IsDog}(x) \wedge \text{Likes}(y, x) \wedge \\ \forall z ((\text{IsDog}(z) \wedge \text{Likes}(y, z)) \Rightarrow z = x)))$$

Semantic parsing example



Semantics of simple sentences

Optional for
the TA
course

$S(\sigma_2(\sigma_1)) \rightarrow NP(\sigma_1) VP(\sigma_2)$

$VP(\sigma_3(\sigma_4)) \rightarrow V(\sigma_3) NP(\sigma_4)$

$NP(\sigma) \rightarrow PN(\sigma)$

$PN(\text{John}) \rightarrow \text{John}$

$PN(\text{Mary}) \rightarrow \text{Mary}$

$V(\lambda y \lambda x \text{ Loves}(x, y)) \rightarrow \text{loves}$

Compositional semantics:
The semantics of each
syntactic constituent is a
function of the semantics of
its sub-constituents.

$(\lambda y \lambda x \text{ Loves}(x, y))(\text{Mary}) \equiv \lambda x \text{ Loves}(x, \text{Mary})$

σ_3 σ_4 $\sigma_3(\sigma_4)$

$(\lambda x \text{ Loves}(x, \text{Mary}))(\text{John}) \equiv \text{Loves}(\text{John}, \text{Mary})$

σ_2 σ_1 $\sigma_2(\sigma_1)$

Corresponding DCG

Optional for
the TA
course

$s(\text{Predicate}) \rightarrow np(X1), vp(X, \text{Predicate}), \{X1 = X\}.$

$vp(X, \text{Predicate}) \rightarrow v(Y, X, \text{Predicate}), np(Y1), \{Y = Y1\}.$

$np(\text{Sem}) \rightarrow pn(\text{Sem}).$

We require the meaning representation Y1 of the np (e.g., john) to be unified with the second argument of the logical predicate of the verb (the Y of loves(X,Y)). This causes the representation of the np to be copied into the predicate of the verb (e.g., loves(X,Y) becomes loves(X, john)).

$pn(\text{john}) \rightarrow [\text{john}].$

$pn(\text{mary}) \rightarrow [\text{mary}].$

The three arguments of v stand for the $\lambda y \lambda x \text{ Loves}(x, y)$ of the previous slide.

$v(Y, X, \text{loves}(X, Y)) \rightarrow [\text{loves}].$

Handling quantifiers

Optional for
the TA
course

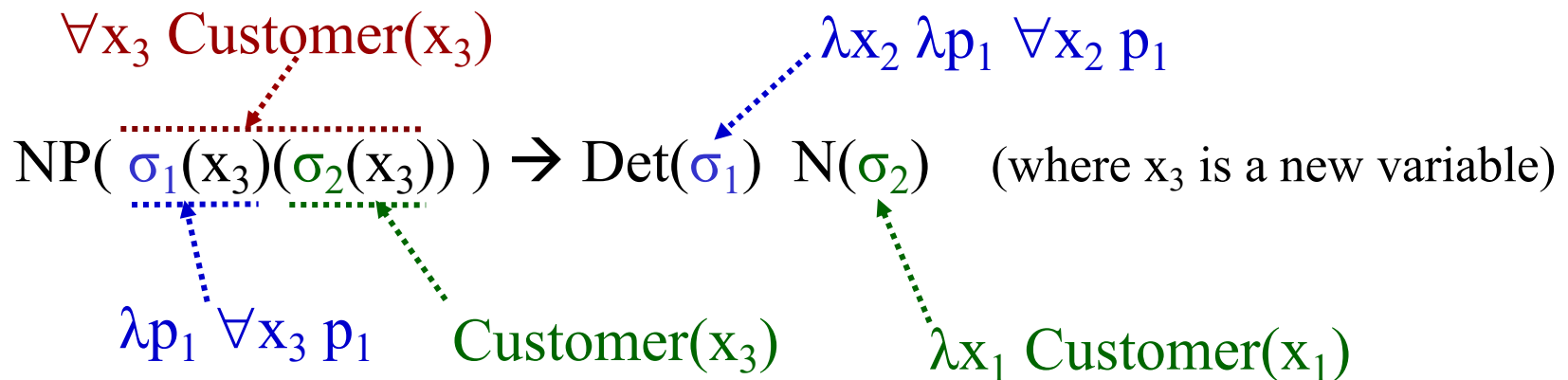
- We extend the grammar to handle sentences like:
 - “I want a flight from Athens to Thessaloniki.”
 - “A customer prefers a flight to Herakleion.”
- We will initially produce formulae with **quantifiers inside the arguments** of the predicates:
 - “Every customer wants a flight.” will initially become:
Wants($\forall x$ Client(x) , $\exists y$ Flight(y))
 - Not allowed in FOPL, but makes semantic parsing easier.
 - Also allows us to produce a single formula for sentences with **ambiguous quantifiers** (see below).
 - A **post-processing stage** will fix the formulae.

Semantics of nouns and determiners

$N(\lambda x_1 \text{ Customer}(x_1)) \rightarrow \text{customer}$

$\text{Det}(\lambda x_2 \lambda p_1 \forall x_2 p_1) \rightarrow \text{every}$

Optional for
the TA
course



Similarly:

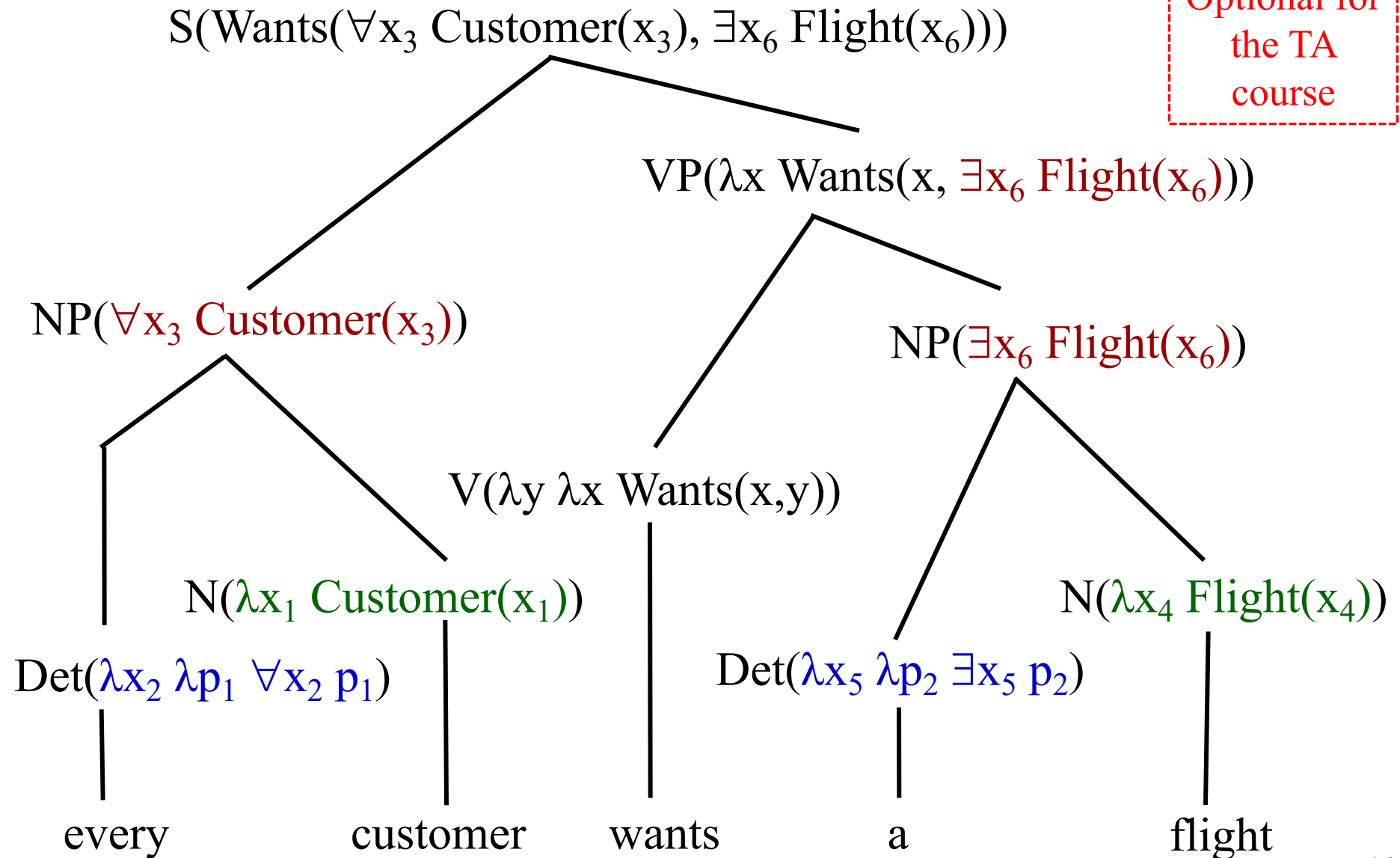
$N(\lambda x_4 \text{ Flight}(x_4)) \rightarrow \text{flight}$

$\text{Det}(\lambda x_5 \lambda p_2 \exists x_5 p_2) \rightarrow \text{a}$

The other rules for verbs, VP, S remain unchanged.

Computing the semantics of a sentence

Optional for
the TA
course



Corresponding DCG rules

Optional for
the TA
course

$n(X1, \text{customer}(X1)) \rightarrow [\text{customer}]$.

$n(X4, \text{flight}(X4)) \rightarrow [\text{flight}]$.

$\text{det}(X2, P1, \text{forall}(X2, P1)) \rightarrow [\text{every}]$.

$\text{det}(X5, P2, \text{forsome}(X5, P2)) \rightarrow [a]$.

The NewX corresponds to the new x_3 variable of the original rules.

$\text{np}(\text{SemNP}) \rightarrow \text{det}(\text{XDet}, P, \text{SemDet}), n(\text{XN}, \text{SemN}),$
 $\{\text{XDet} = \text{NewX}, \text{XN} = \text{NewX}, P = \text{SemN}, \text{SemNP} = \text{SemDet}\}.$

or more briefly:

$\text{np}(\text{SemDet}) \rightarrow \text{det}(X, \text{SemN}, \text{SemDet}), n(X, \text{SemN}).$

Converting to true FOPL

- A **post-processor** produces all the possible FOPL formulae from the intermediate formula.
 - Wants($\forall x$ Customer(x), $\exists y$ Flight(y)) becomes:
 - $\forall x$ Customer(x) $\rightarrow$ ($\exists y$ Flight(y) $\wedge$ Wants(x, y)) **or**:
 - $\exists y$ Flight(y) $\wedge$ ($\forall x$ Customer(x) $\rightarrow$ Wants(x, y))
- **Cut a quantifier** (and its predicate) from the intermediate formula **leaving only its variable**:
 - E.g., Wants(x , $\exists y$ Flight(y))
- **Paste the quantifier** and its predicate at the **beginning of the formula**.
 - **Connect with $\Rightarrow$** if the quantifier is a $\forall$.
 - **Connect with $\wedge$** if the quantifier is a $\exists$.
 - $\forall x$ Customer(x) $\Rightarrow$ Wants(x , $\exists y$ Flight(y))

Converting to true FOPL – cont.

Optional for
the TA
course

- Repeat for the **other** quantifier:
 - $\forall x \text{ Customer}(x) \Rightarrow \text{Wants}(x, y)$
 - $\exists y \text{ Flight}(y) \wedge (\forall x \text{ Customer}(x) \Rightarrow \text{Wants}(x, y))$
 - There is a **single** (the same for all customers) **flight** that they all want.
- Cutting the existential quantifier first, leads to another reading:
 - $\forall x \text{ Customer}(x) \Rightarrow (\exists y \text{ Flight}(y) \wedge \text{Wants}(x, y))$
 - For every customer, there is a **possibly different flight** that the customer wants.
- The intermediate formula may also contain **pseudo-quantifiers representing referring expressions**.
 - “**The customer** wants a flight.”
 - $\text{Wants}(\text{The.x Customer}(x), \exists y \text{ Flight}(y))$
 - The pseudo-quantifiers are replaced by appropriate expressions using algorithms that resolve referring expressions.

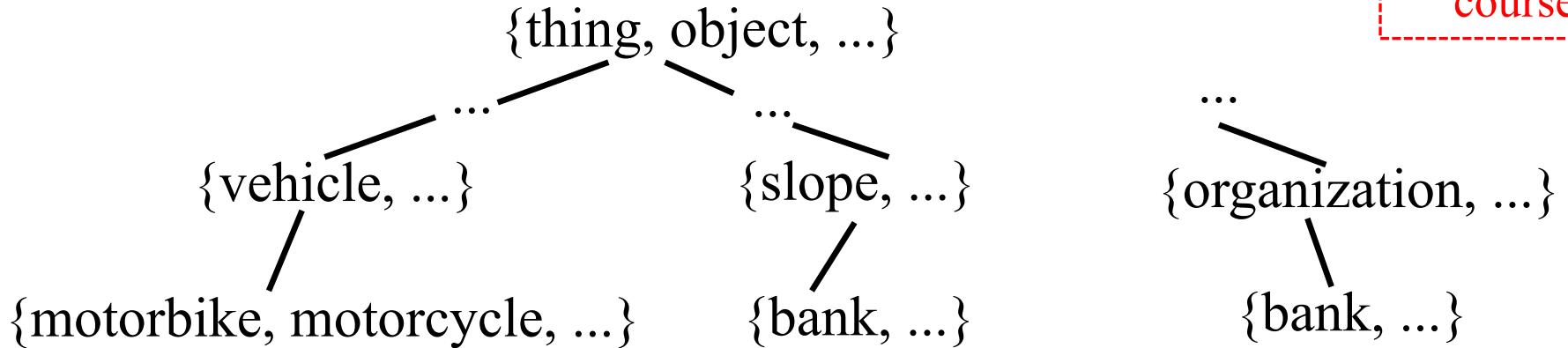
Lexical semantic relations

Optional for
the TA
course

- **Homonyms:** same spelling, different meanings.
 - E.g., financial “bank” and “bank” of a river.
 - E.g., “letter” of the alphabet and “letter” that you post.
- **Synonyms:** different words, but can be used with (approximately) the same meaning.
 - E.g., “motorbike” and “motorcycle”, “lift” and “elevator”.
- **Hypernym – hyponym:** broader – narrower meaning.
 - E.g., “vehicle” – “car”, “organization” – “company”.
- **Antonyms:** opposite meanings.
 - E.g., “tall” – “short”, “large” – “small”.
- See J&M for more kinds of lexical semantic relations.

WordNet (<http://wordnet.princeton.edu/>)

Optional for
the TA
course



- Every **sense** is represented by a **set of synonyms** (synset) that can have that sense.
- **Hypernym – hyponym** hierarch per part of speech (nouns, adjectives, verbs, adverbs).
- Many **other relations** also included (see J&M).
 - E.g., **meronyms** (a “wheel” is part of a “bicycle”).
- Initially for **English**, now for **many languages**.

Word to search for:

 Display Options: (Select option to change)

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations

 Optional for
the TA
course

Noun

- [S: \(n\) bank](#) (sloping land (especially the slope beside a body of water)) *"they pulled the canoe up on the bank"; "he sat on the bank of the river and watched the currents"*
 - [direct hyponym](#) / [full hyponym](#)
 - [S: \(n\) riverbank, riverside](#) (the bank of a river)
 - [S: \(n\) waterside](#) (land bordering a body of water)
 - [direct hypernym](#) / [inherited hypernym](#) / [sister term](#)
 - [S: \(n\) slope, incline, side](#) (an elevated geological formation) *"he climbed the steep slope"; "the house was built on the side of a mountain"*
 - [derivationally related form](#)
- [S: \(n\) depository financial institution, bank, banking concern, banking company](#) (a financial institution that accepts deposits and channels the money into lending activities) *"he cashed a check at the bank"; "that bank holds the mortgage on my home"*
 - [direct hyponym](#) / [full hyponym](#)
 - [member holonym](#)
 - [direct hypernym](#) / [inherited hypernym](#) / [sister term](#)
 - [S: \(n\) financial institution, financial organization, financial organisation](#) (an institution (public or private) that collects funds (from the public or other institutions) and invests them in financial assets)
 - [derivationally related form](#)
- [S: \(n\) bank](#) (a long ridge or pile) *"a huge bank of earth"*
- [S: \(n\) bank](#) (an arrangement of similar objects in a row or in tiers) *"he operated a bank of*

Information extraction

Dec. 3, 2015: Important news from General Company Hellas, the largest Greek construction company. Yesterday GCH announced it bought 42% of Small Company Ltd, a British company that specializes in iron constructions.



buyer	GCH
bought	SCL
share	0.42
announcement date	2/12/2015

Information extraction: first stages

<s> <date norm="3/12/2015"> Dec. 3, 2015 </date> : </s>

<s> Important news from <company id="GCH"> General Company Hellas </company>, the largest Greek construction company. </s>

<s> <date norm="2/12/2015"> Yesterday </date> <company id="GCH"> GCH </company> announced it <verb base="buy"> bought </verb> <percent norm="0.42"> 42% </percent> of <company id="SCL"> Small Company Ltd </company>, a British company that specializes in iron constructions. </s>

Information extraction: first stages

- **Preprocessing:**
 - **Tokenization, sentence splitting, markup processing** etc.
- **Morphological analysis:**
 - **Part-of-speech tagging, lemmatization** etc.
- **Dates, amounts, percentages** etc.
 - Including **normalization** (e.g., “Dec. 3, 2015”, 3/12/2015), often using simple regular expressions or grammars.
- **Named entity recognition** (e.g., persons, companies, locations, products, diseases, genes).
 - **Sequence labeling** algorithms (e.g., biLSTM + CRF layer).
 - **Features: embeddings of words, token shapes, POS tags, prefixes/suffixes**, extra features from **gazetteers** (e.g., known company names, cities, surnames, person titles), ...

A neural NE recognizer

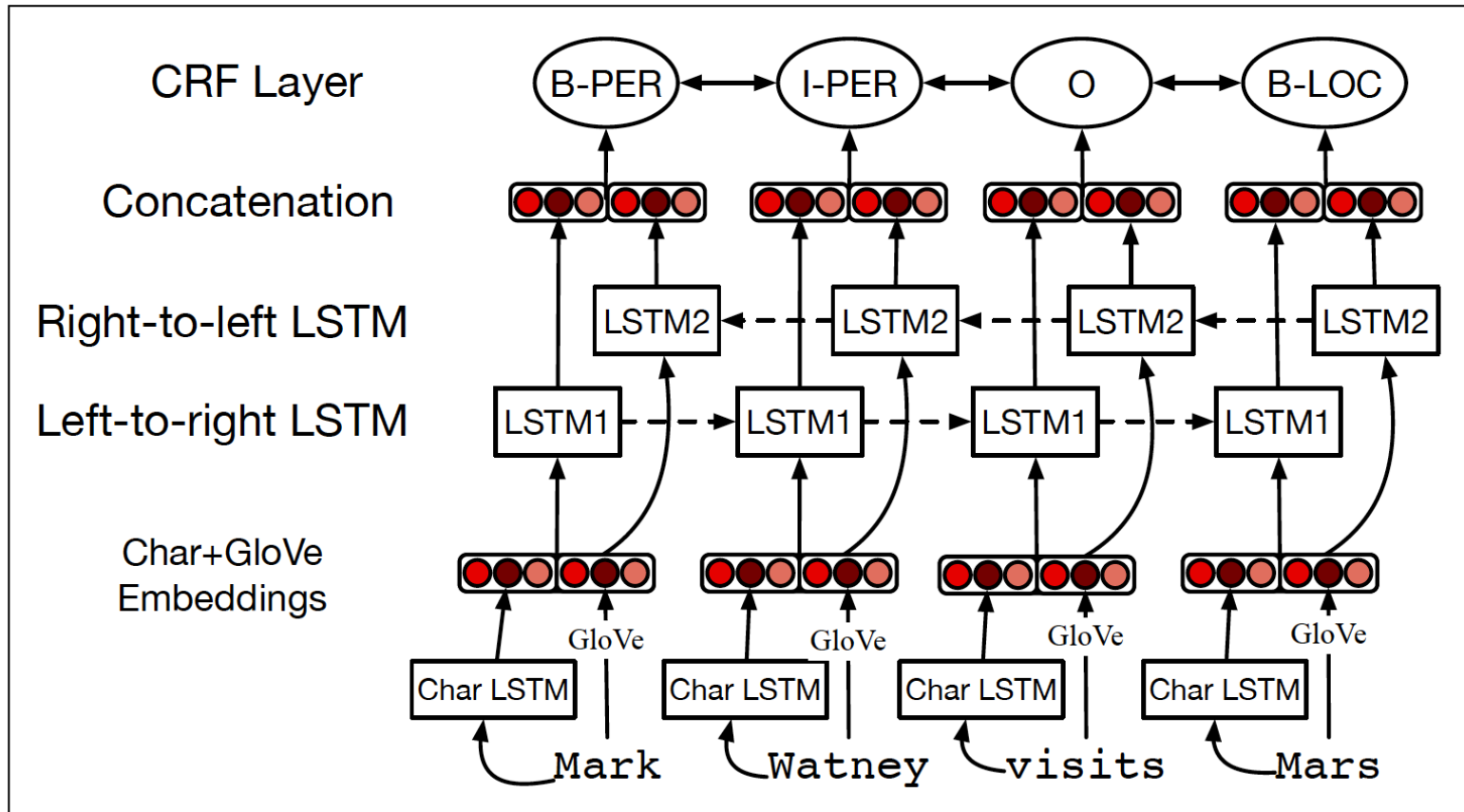


Figure 18.8 Putting it all together: character embeddings and words together in a bi-LSTM sequence model. After Lample et al. (2016).

Figure from Jurafsky and Martin's "Speech and Language Processing",
3rd edition (in preparation). <https://web.stanford.edu/~jurafsky/slp3/>

Information extraction: next stages

- **Coreference resolution.**

- “**She** also said that...”. “**The company** also announced that...”
- Including **name matching** (e.g., “General Company Hellas”, “GCH”, “Mr. George Papandreou”, “Papandreou”) and possibly **linking entity mentions to ontology concepts** (ids).

- **Relation extraction:**

- In the simplest case, using **manually crafted rules**.
- E.g., Acquisition(**buyer:C1**, bought:C2, **share:P**) → *
Company(id:C1) * Verb(base:buy/acquire/obtain) *
Percent(norm:P) * **Company(id:C2)** *
- Or using **supervised machine learning**: learn to **predict the relation** (if any) between each **pair of named entities** (that do not exceed a maximum distance). **Classification** problem. One **class per relation type** (plus ‘none’).

Rule-based relation extraction

<s> Important news from <company id="GCH"> General Company Hellas </company>, the largest Greek construction company. </s>

<s> <date norm="2/12/2015"> Yesterday </date> **<company id="GCH"> GCH </company>** announced that it **<verb base="buy"> bought</verb>** **<percent norm="0.42"> 42% </percent>** of **<company id="SCL"> Small Company Ltd </company>**, a British company that specializes in iron constructions. </s>

Acquisition(**buyer:C1**, **bought:C2**, **share:P**) → *
Company(id:C1) * **Verb(base:buy/acquire/obtain)** *
Percent(norm:P) * **Company(id:C2)** *

Acquisition(**buyer:GCH**, **bought:SCL**, **share:0.42**)

Relation extraction via supervised learning

<s> <company id="GCH"> General Company Hellas </company>
<verb base="buy"> bought </verb> <percent norm="0.42"> 42%
 </percent> of <company id="SCL"> Small Company Ltd
</company>. </s> <s> <company id="LCL"> Large Company Ltd
</company> had also <verb base="buy"> bought </verb> shares of
<company id="SCL"> Small </company> <date norm="Y2007">
 last year </date> . </s>

- E.g., for **acquisition** relations:
 - Consider **company-company** and **person-company** pairs (up to a maximum distance).
 - **Classes** (for each pair): **negative**, **positive** (or type of relation, e.g., acquisition, merger).

Relation extraction via supervised learning

- **Possible features for each pair of named entities:**

...before... Bank of England ...between... LanghTech ...after...

- **Embeddings** of words, n-grams, POS tags, token shapes, etc. of the two **named entities**, of the words **between** them, **before** the first named entity, **after** the second named entity.
 - Possibly using, for example, the **centroid** of the **word embeddings** of **each named entity**, the **centroid** of words **before**, **between**, **after**.
 - Or using a **biLSTM** to obtain a **single vector** for each named entity, and for the words **before**, **between**, **after** the two named entities.
- **Types** (or embeddings of types) of the two **named entities** (e.g., person, organization, location).
- **Head words** (or their embeddings) of the named entities (e.g., “the Queen of England”, “Bank of...”) from dependency trees.
- **Distance** between the two **named entities** (counted in **words**, **parse tree path length**, intervening **other named entities**).

Joint NE and relation extraction

A binary classifier scores the possible heads & relations of each word (e.g., Smith worksFor Center)

Embedding of NER label (e.g., I-PER)

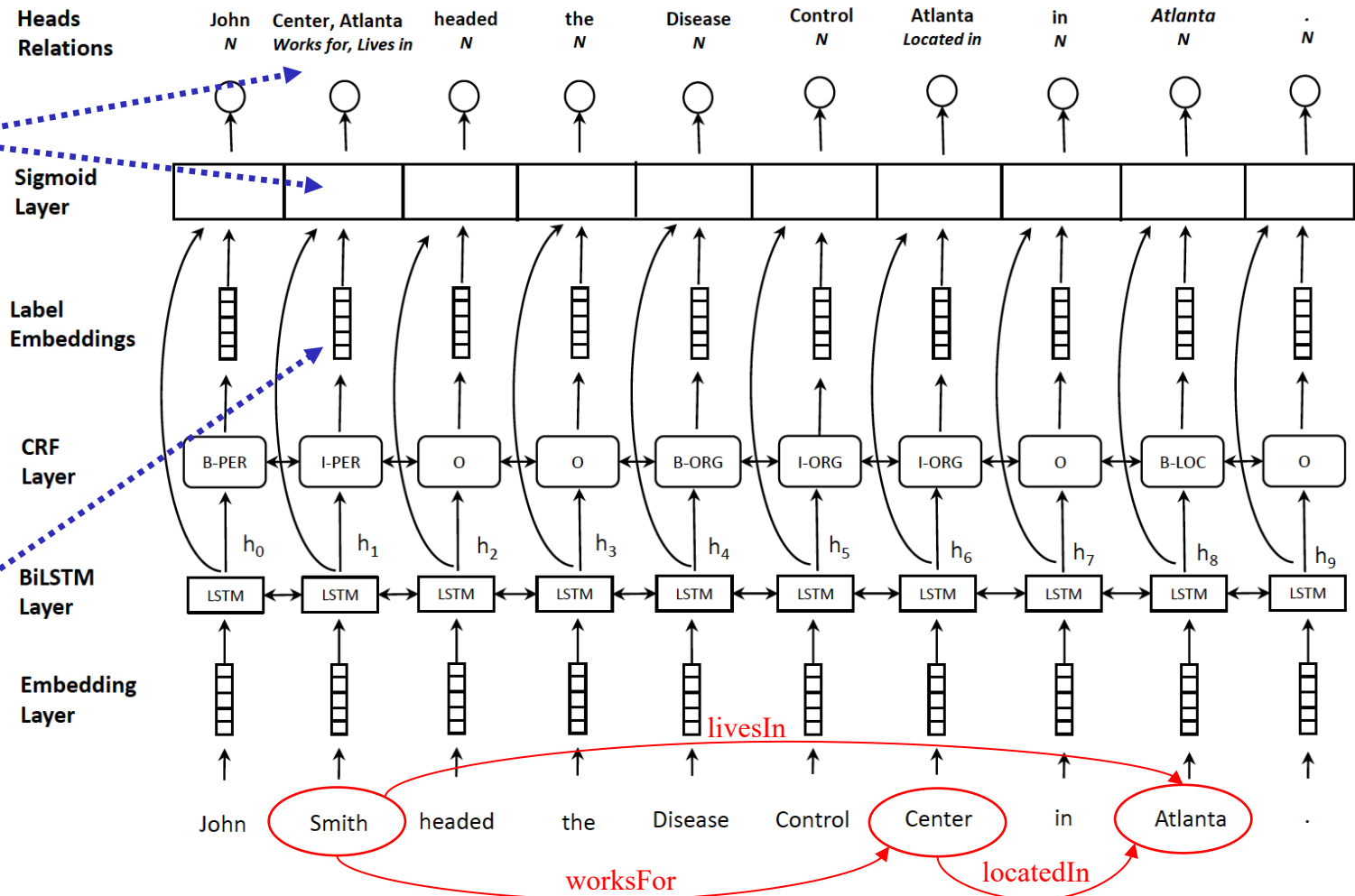
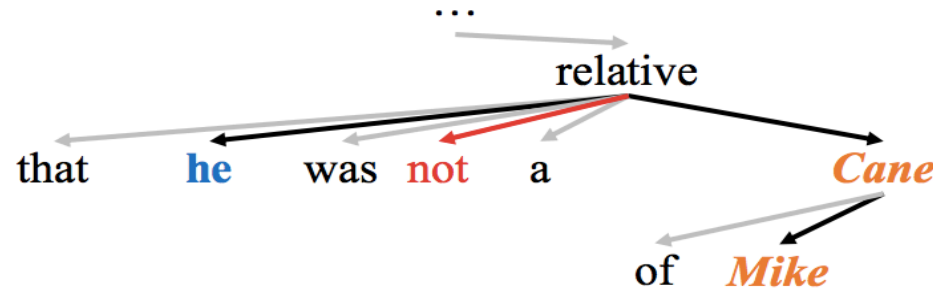


Figure from G. Bekoulis et al., “Joint entity recognition and relation extraction as a multi-head selection problem”, *Expert Systems with Applications* 114, pp. 34–45, 2018. See also G. Bekoulis et al., “Adversarial training for multi-context joint entity and relation extraction”, EMNLP 2018, <https://www.aclweb.org/anthology/D18-1307/>

Relation extraction with graph CNNs

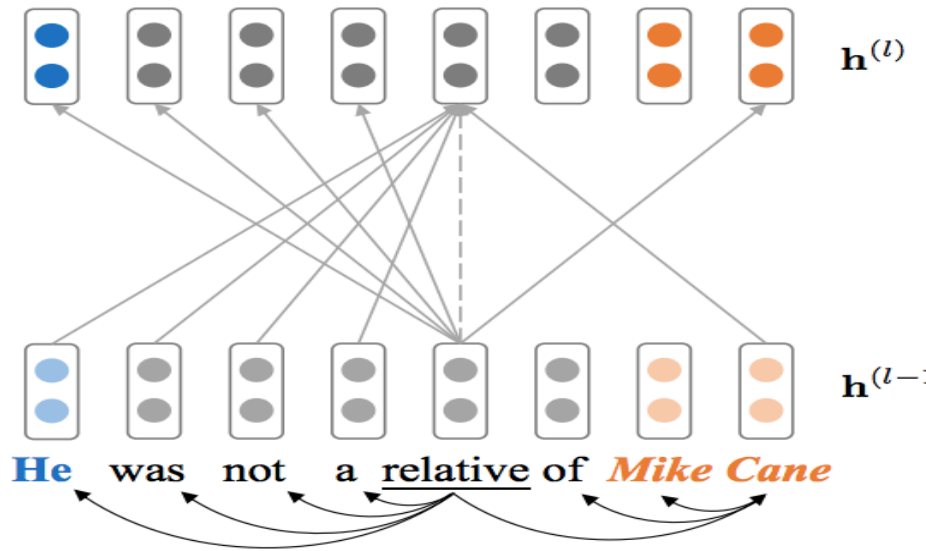
I had an e-mail exchange with Benjamin Cane of Popular Mechanics which showed that **he** was not a relative of **Mike Cane**.



Prediction from dependency path: *per:other_family*
Gold label: *no_relation*

- We are **given** the **dependency tree** and the spans (positions in text and tree) of two **named entities** (S, O).
 - Here “he” and “Mike Cane”.
- We need to **predict** the **type** of their **relation** (if any).

Relation extraction with graph CNNs (cont.)



Matrix (convolution kernel) and bias term of level l .

$$h_i^{(l)} = \sigma \left(\sum_{j=1}^n A_{ij} W^{(l)} h_j^{(l-1)} + b^{(l)} \right)$$

$A_{i,j} = 1$ if word w_i is connected to word w_j or $i = j$, otherwise 0.

- We build **representations** of the **words** at **levels** 1, 2,
 - At **level one**, each **word** w_i is represented by its **embedding**.
 - At **level** l , the representation $h_i^{(l)}$ of each **word** w_i is a **combination** of the **lower-level representations** $h_j^{(l-1)}$ of the words w_j it is **connected** to in the **dependency tree** (incl. itself).

Relation extraction with graph CNNs (cont.)

Max-pooling of top-level representations of the words in entity S, all the sentence, entity O.

$$h_{\text{final}} = \text{FFNN}([h_{\text{sent}}; h_s; h_o])$$

Fed to a linear layer with softmax to predict the relation type (if any) between the two entities (S, O).

Concatenation

Pooling

GCN Output
 $\mathbf{h}^{(L)}$

GCN Input
 $\mathbf{h}^{(0)}$

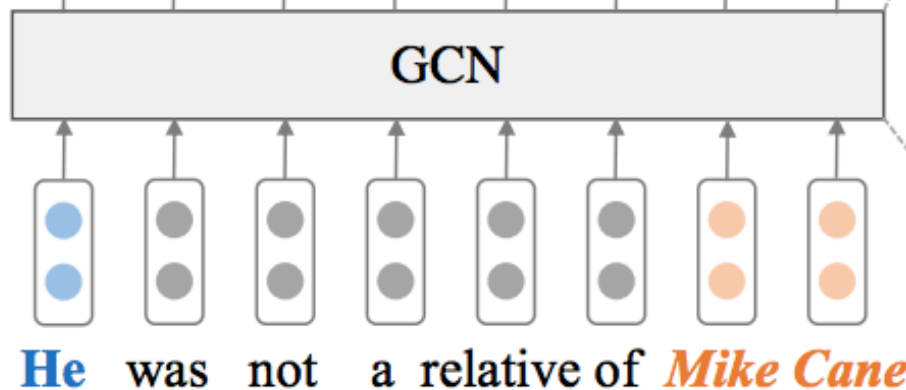


Figure from Y. Zhang, P. Qi, C.D. Manning, "Graph Convolution over Pruned Dependency Trees Improves Relation Extraction", EMNLP 2018. <http://aclweb.org/anthology/D18-1244>

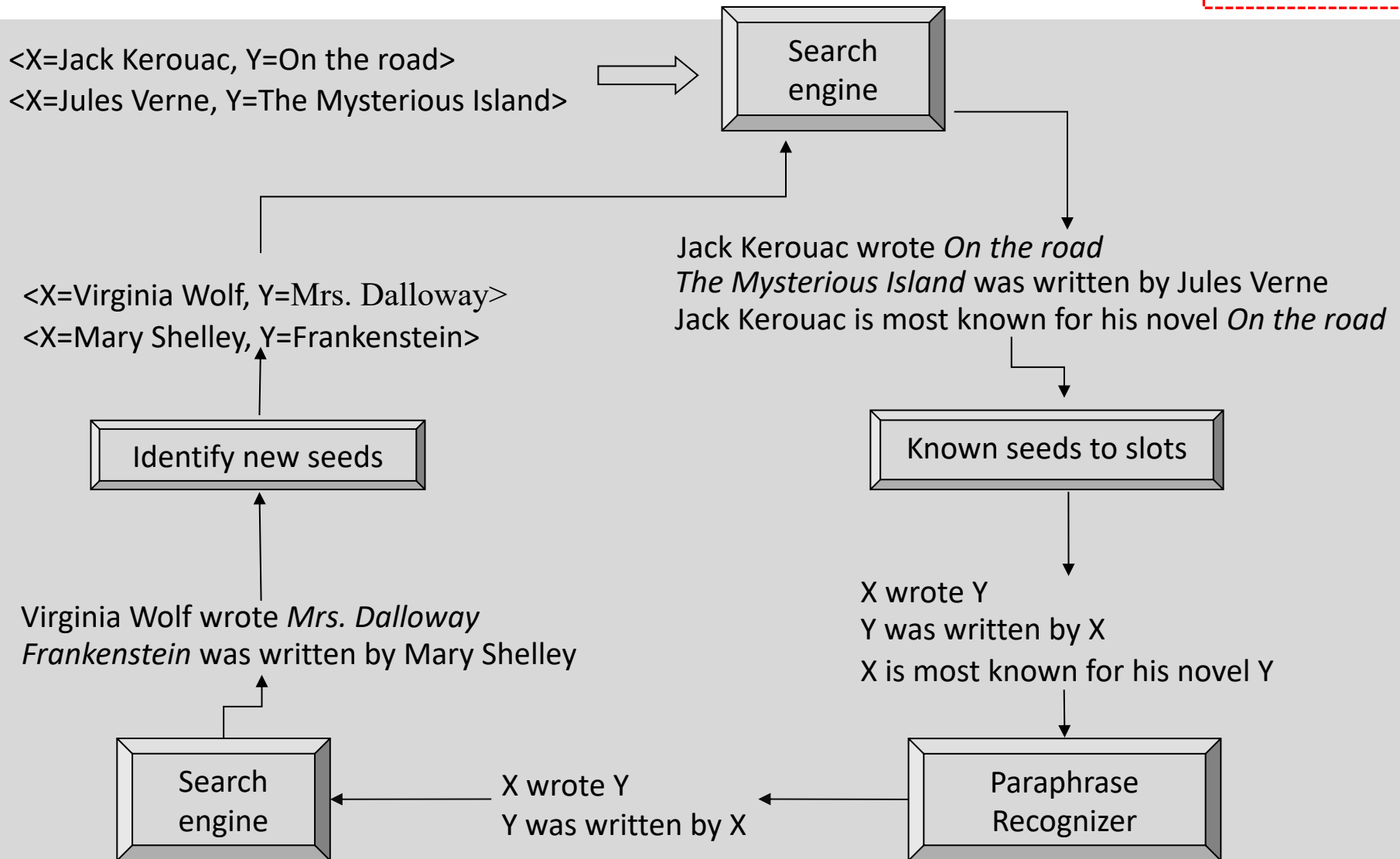
Evaluating relation extraction

- We can use **precision, recall, F-measure** again.
 - If we have **texts manually annotated** with the **correct relations**, we can count how many of the **correct relations** were **extracted (true positives)**, how many were **not extracted (false negatives)**, how many **wrong relations** were **extracted (false positives)** etc.
 - If we have a **database** with all the **correct entity pairs per relation type**, we can count how many of the **correct entity pairs** were **extracted (true positives)**, how many were **not extracted (false negatives)**, how many **wrong entity pairs** were **extracted (false positives)** etc.

Extra optional slides.

(almost) Unsupervised relation extraction

Optional
reading



(almost) Unsupervised relation extraction

Optional
reading

- **For each relation type, perform bootstrapping:**
 - Start with **seed entity pairs** for which the relation holds.
 - **Retrieve sentences** that **contain** the **seed entity pairs**.
 - **Construct patterns** from the **retrieved sentences**, e.g., regular expression patterns or patterns operating on parse trees.
 - Retrieve **sentences** that **match** the **patterns**.
 - Extract **new entity pairs** from the **retrieved sentences** etc.
- **Semantic drift** problem:
 - Soon we start obtaining **entity pairs** and **patterns** that are very **general** or **irrelevant** (e.g., “X is known for Y”).
 - A “**human in the loop**” is needed to **filter** the **entity pairs** and **patterns**, or **measures** to automatically **score them**, or **classifiers** to filter them (possibly trained on human decisions).

Scoring new patterns

Optional
reading

- A good **candidate pattern** p will have **high precision**, but will also **fire frequently** in a corpus (high recall).
 - Patterns with **low precision** are **unreliable**.
 - Patterns that **rarely fire** will also **not help** much.

$$\text{Conf}(p) = \frac{\text{hits}(p)}{\text{finds}(p)} \cdot \log \text{finds}(p)$$

How many
**known correct
entity pairs** does
 p **extract** from
the corpus?

How many **entity
pairs** does p **extract**
from the corpus?

Scoring new entity pairs

Optional
reading

- Assume an **entity pair** r is **wrong** if and only if **all the patterns** $p \in P$ that **extract (support)** r are **mistaken**.
 - Assume that the **mistakes of the patterns** are **independent**.
 - And that the **confidence** $\text{Conf}(p)$ of each **pattern** p is (almost) a **probability**.

“Probability” that **entity pair** r is **correctly extracted** (not all of the patterns that extract it are mistaken).

“Probability” that **all the patterns** that **extract** r are **mistaken**.

$$\text{Conf}(r) = 1 - \prod_{p: p \text{ extracts } r} (1 - \text{Conf}(p))$$

“Probability” that a **pattern** p that **extracts** r is **mistaken**.

Syntax of FOPL

Optional
reading

formula $\rightarrow$ *atomic_formula*

| (*formula* *connective* *formula*)

| *quantifier* *variable* *formula*

| $\neg$ *formula*

atomic_formula $\rightarrow$ *relation_symbol*(*term*, ...) | *term* = *term*

term $\rightarrow$ *constant* | *variable* |

function_symbol(*term*, ...)

connective $\rightarrow$ $\wedge$ | $\vee$ | $\Rightarrow$ | $\Leftrightarrow$

quantifier $\rightarrow$ $\forall$ | $\exists$

constant $\rightarrow$ A | X₁ | John | Mary | ...

variable $\rightarrow$ a | x | s | ...

relation_symbol $\rightarrow$ IsFatherOf | HasColor | IsKing | ...

function_symbol $\rightarrow$ FatherOf | LeftLeg | ...

Representing events

Optional
reading

I ate.

$\text{Eating}_1(\text{Speaker})$

I ate a souvlaki.

$\exists x (\text{IsSouvlaki}(x) \wedge \text{Eating}_2(\text{Speaker}, x))$

I ate a souvlaki at my office.

$\exists x (\text{IsSouvlaki}(x) \wedge \text{Eating}_3(\text{Speaker}, x, \text{OfficeOf}(\text{Speaker})))$

I ate a souvlaki at my office yesterday.

$\exists x (\text{IsSouvlaki}(x) \wedge$
 $\text{Eating}_4(\text{Speaker}, x, \text{OfficeOf}(\text{Speaker}), \text{Yesterday}))$

- If we use separate $\text{Eating}_1(\dots)$, $\text{Eating}_2(\dots)$, $\text{Eating}_3(\dots)$ etc. predicates, we need meaning postulates stating, for example, that when $\text{Eating}_3(\dots)$ happens then $\text{Eating}_2(\dots)$ also happens.

Representing events – cont.

Optional
reading

I ate.

$$\exists x \exists y \exists z \text{ Eating}(\text{Speaker}, x, y, z)$$

I ate a souvlaki.

$$\exists x \exists y \exists z (\text{IsSouvlaki}(x) \wedge \text{Eating}(\text{Speaker}, x, y, z))$$

I ate a souvlaki at my office.

$$\exists x \exists z (\text{IsSouvlaki}(x) \wedge \text{Eating}(\text{Speaker}, x, \text{OfficeOf}(\text{Speaker}), z))$$

I ate a souvlaki at my office yesterday.

$$\exists x (\text{IsSouvlaki}(x) \wedge \text{Eating}(\text{Speaker}, x, \text{OfficeOf}(\text{Speaker}), \text{Yesterday}))$$

- How many arguments does $\text{Eating}(\dots)$ need?
- What about “I quickly ate a souvlaki at my office yesterday”?
- What about “I quickly ate a souvlaki at my office yesterday before leaving”?

Event variables (Davidsonian semantics)

Optional
reading

I ate.

$$\exists e (\text{Eating}(e) \wedge \text{Eater}(e, \text{Speaker}))$$

I ate a souvlaki.

$$\exists e \exists x (\text{Eating}(e) \wedge \text{IsSouvlaki}(x) \wedge \text{Eater}(e, \text{Speaker}) \wedge \\ \text{Eaten}(e, x))$$

I ate a souvlaki at my office.

$$\exists e \exists x (\text{Eating}(e) \wedge \text{IsSouvlaki}(x) \wedge \text{Eater}(e, \text{Speaker}) \wedge \\ \text{Eaten}(e, x) \wedge \text{Location}(e, \text{OfficeOf}(\text{Speaker})))$$

I quickly ate a souvlaki at my office before leaving.

$$\exists e_1 \exists e_2 \exists x \exists i_1 \exists i_2 (\text{Eating}(e_1) \wedge \text{IsSouvlaki}(x) \wedge \text{Eaten}(e_1, x) \wedge \\ \text{Eater}(e_1, \text{Speaker}) \wedge \text{Location}(e_1, \text{OfficeOf}(\text{Speaker})) \wedge \\ \text{Speed}(e_1, \text{Fast}) \wedge \text{Leaving}(e_2) \wedge \text{Leaver}(e_2, \text{Speaker}) \wedge \\ \text{IntervalOf}(e_1, i_1) \wedge \text{IntervalOf}(e_2, i_2) \wedge \text{Before}(\text{End}(i_1), \text{Start}(i_2)))$$

Thematic roles

Optional
reading

I quickly ate a souvlaki at my office before leaving.

$$\begin{aligned} &\exists e_1 \exists e_2 \exists x \exists i_1 \exists i_2 (Eating(e_1) \wedge IsSouvlaki(x) \wedge Eaten(e_1, x) \wedge \\ &Eater(e_1, Speaker) \wedge Location(e_1, OfficeOf(Speaker)) \wedge \\ &Speed(e_1, Fast) \wedge Leaving(e_2) \wedge Leaver(e_2, Speaker) \wedge \\ &IntervalOf(e_1, i_1) \wedge IntervalOf(e_2, i_2) \wedge Before(End(i_1), Start(i_2))) \end{aligned}$$

- Depending on the **event type**, different **roles** are available.
 - **Every event** has an **IntervalOf** role.
 - An **Eating** event may also have **Eaten** and **Eater** roles.
 - A **Leaving** event may also have a **Leaver** role.
- We need a **taxonomy of event types**, which will define the possible **event types** and their **roles**.

Frame Index

https://framenet2.icsi.berkeley.edu/

Frame Index

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Abandonment
Abounding with
Absorb heat
Abundance
Abusing
Access scenario
Accompaniment
Accomplishment
Accoutrements
Accuracy
Achieving first
Active substance
Activity
Activity abandon
Activity done st
Activity finish
Activity ongoing
Activity pause
Activity paused
Activity prepare
Activity ready s
Activity resume
Activity start
Activity stop
Actually occurri
Addiction
Adding up

Abandonment

[Lexical Unit Index](#)

Definition:

An **Agent** leaves behind a **Theme** effectively rendering it no longer within their control or of the normal security as one's property.

Carolyn **ABANDONED** **her car** and jumped on a red double decker bus.

Perhaps **he** **LEFT** **the key** in the ignition

ABANDONMENT of a child is considered to be a serious crime in many jurisdictions.

There are also metaphorically used examples:

She **LEFT** **her old ways** **behind**.

FEs:

Core:

Agent [Age] The **Agent** is the person who acts to leave behind the **Theme**.

Theme [The] The **Theme** is the entity that is relinquished to no one from the **Agent**'s possession.

Non-Core:

Degree [] The extent to which the **Agent** leaves the **Theme** behind.

framenet2.icsi.berkeley.edu/

Non-Core:

Degree [] The extent to which the **Agent** leaves the **Theme** behind.

Depictive [] The FE **Depictive** describes the **Agent** during the abandoning event.

Duration [Dur] For what expanse of time the **Agent** has given up the **Theme**.

Event_descriptor [evdesc] A description of the event as a whole.

Explanation [] **Explanation** denotes a proposition from which the act of abandonment logically follows.

Manner [Man] The style in which the **Agent** gives up the **Theme**.

Means [] An action performed by the **Agent** that accomplishes the action indicated by the target.

Place [Pla] The location where the **Agent** gives up the **Theme**.

Purpose [] The purpose for which the **Agent** abandons the **Theme**.

Time [Tim] When the **Agent** gives up the **Theme**.

Frame-frame Relations:

Inherits from: [Intentionally affect](#)

FrameNet (<https://framenet.icsi.berkeley.edu/>)

Optional reading

- Particular trigger words **activate frames**, which define **thematic roles** (frame elements).
 - “... an *increase* [INIT_VALUE from 20%] [FINAL_VALUE to 27%]...”
 - “... *fell* [FINAL_VALUE to 27%]...”
 - Here both trigger words activate the same frame.
- **FrameNet** provides a rich collection of **frames**, **trigger words**, **roles**, **inheritance** from more general to more specific frames etc.
 - Useful, for example, in **information extraction**.
 - **Semantic role labeling**: methods that “fill” the roles of active frames in each sentence, usually by employing machine learning (see J&M).

Selectional restrictions

Optional
reading

I saw the [doctor [with the white coat]].

$$\exists e_1 \exists x_1 \exists x_2 \exists i_1 (\text{Seeing}(e_1) \wedge \text{IsDoctor}(e_1, x_1) \wedge \text{IsCoat}(e_1, x_2) \wedge \text{IsWhite}(e_1, x_2) \wedge \text{Agent}(e_1, \text{Speaker}) \wedge \text{Seen}(e_1, x_1) \wedge \text{Wearing}(e_1, x_1, x_2) \wedge \text{IntervalOf}(e_1, i_1) \wedge \text{Before}(\text{End}(i_1), \text{Now})))$$

? *I saw [the doctor] [with the white coat].*

$$\exists e_1 \exists x_1 \exists x_2 \exists i_1 (\text{Seeing}(e_1) \wedge \text{IsDoctor}(e_1, x_1) \wedge \text{IsCoat}(e_1, x_2) \wedge \text{IsWhite}(e_1, x_2) \wedge \text{Agent}(e_1, \text{Speaker}) \wedge \text{Seen}(e_1, x_1) \wedge \text{ObservationInstrument}(e_1, x_2) \wedge \text{IntervalOf}(e_1, i_1) \wedge \text{Before}(\text{End}(i_1), \text{Now})))$$

The 2nd reading can be ruled out via **logical inference**, if we have a sufficiently rich **knowledge base**. Difficult...

$$\forall e \forall x (\text{ObservationInstrument}(e, x) \Leftrightarrow (\text{IsEyeGlasses}(e, x) \vee \text{IsBinoculars}(e, x) \vee \dots))$$

Selectional restrictions – cont.

Optional
reading

- Alternatively, simple **selectional restrictions** can be included in the **lexicon** and **grammar**.

$n(\text{sense: } s_{144}) \rightarrow [\text{food}]$. (Assuming that s_{144} is the **synset** for the concept of **food**.)

$n(\text{sense: } s_{138}) \rightarrow [\text{salad}]$. (Assuming s_{138} is a **hyponym** of s_{144} .)

$v(\text{objSense: } s_{144}) \rightarrow [\text{eat}]$. (The argument of v shows that the verb **requires** an **object** with sense s_{144} or **hyponym**.)

$vp \rightarrow v(\text{objSense: } S_1), n(\text{sense: } S_2), \{\text{hypernymOf}(S_1, S_2)\}$.

- Similarly** (exercise...) we can **rule out**:

I saw [the doctor] [with the white coat].

- No parse trees** produced for **readings violating selectional restrictions**.
 - But **WordNet** may not provide exactly the **concepts we need** for some selectional restrictions.
 - Also problems with **negations** (e.g., “Do not drink gasoline.”) or **metaphors** (“Time flies when you’re having fun.”).

Temporal expressions

Optional
reading

- **Some kinds** of temporal expressions:
 - **Temporal locations:** e.g., “on January 18th, 2015”, “in the 5th century BC”.
 - **Relative temporal locations:** e.g., “yesterday”, “two weeks earlier”, “the following two years”. Possibly related to the **publication date** or other **prominent time** (e.g., of an event).
 - **Durations, frequencies:** “the battery lasts for 8 hours”, “payment is due in three days/every January”.
- **Many temporal mechanisms** in languages:
 - **Tense/aspect** of verbs, temporal **adjectives/adverbs**, **clauses**...

Recognizing temporal expressions

- Temporal expressions often contain **trigger words**.
 - E.g., “January”, “week”, “year”, “Sunday”.
 - But there are exceptions too (e.g., “Never on a Sunday”).
- We can use the **same supervised learning** methods as in **named entity recognition** (sequence labeling).
- For simple expressions (e.g., dates, durations) manually crafted **regular expressions** or **grammars** may suffice.
 - Reasonably easy to write for simple temporal expressions.
 - Grammars can be extended to also **normalize** the temporal expressions (e.g., “on January 18, 2015” → “18/1/2015”).

Athens, <TIMEX3 id="t1" type="DATE" value="2015-07-02"
functionInDocument="CREATION_TIME"> July 2,
2015</TIMEX3> : The reduced ticket prices that OA announced
<TIMEX3 id="t2" type="DATE" value="2015-W26"
anchorTimeID="t1"> last week </TIMEX3> forced ...

- **TimeML: Annotation standard** (XML-based) for **temporal expressions, their normalized values, and events.**
 - **TimeBank:** corpus annotated according to TimeML.
 - See J&M for further details.

Recommended reading

- J&M (2nd ed.): chapters 17, 18, 19, 20, 21, 22.
 - Material not covered by the slides is optional.
 - Check also the 3rd edition (in preparation):
<http://web.stanford.edu/~jurafsky/slp3/>
- J. Eisenstein, “Natural Language Processing”, MIT Press, 2019: chapters: 12, 13.
 - Free draft available at <https://github.com/jacobeisenstein/gt-nlp-class/blob/master/notes/eisenstein-nlp-notes.pdf>
 - Material not covered by the slides is optional.

